

Lecture 6: Concentration Bounds for Sum of Independent Matrices

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Motivation: From Scalars to Matrices

Theorem 1 (Bernstein's inequality)

Let $X_1, \dots, X_n$ be independent zero-mean random variables with $|X_i| \leq M$ a.s. Then, for all $t > 0$,

$$\Pr\left(\sum_{i=1}^n X_i > t\right) \leq \exp\left(-\frac{\frac{1}{2}t^2}{\sum_{i=1}^n \mathbb{E}[X_i^2] + \frac{1}{3}Mt}\right).$$

Towards random matrices

In random matrix theory, one is similarly interested in **sums of independent random matrices**.

In the following, we will see a **matrix generalization of Bernstein's inequality**, focusing on the case of **independent, random, Hermitian (inherently square) matrices** (similar considerations can be extended to rectangular matrices).

Square Matrices and Eigenvalues

Focus of our analysis

Since we are dealing with **square random matrices**, we study their **eigenvalues**.

- We aim to derive an **upper bound for the maximum eigenvalue**.
- A symmetric argument can yield a **lower bound for the minimum eigenvalue** (see Proposition 3.2.1 in J.A. Tropp, [An Introduction to Matrix Concentration Inequalities](#)).

Analytic tool: the Laplace transform method

This method leverages the **moment generating function (mgf)** and the **cumulant generating function (cgf)** of a random matrix $\mathbf{X}$.

Moment and Cumulant Generating Functions

Definition 1 (MGF and CGF of a random matrix $\mathbf{X}$)

- **Moment generating function:**

$$M_{\mathbf{X}}(\theta) := \mathbb{E}[e^{\theta \mathbf{X}}].$$

- **Cumulant generating function:**

$$K_{\mathbf{X}}(\theta) := \log \mathbb{E}[e^{\theta \mathbf{X}}].$$

Example: Moment Generating Function of a Gaussian

Let $X \sim \mathcal{N}(0, \sigma^2)$. We compute its **moment generating function (mgf)**:

$$M_X(\theta) = \mathbb{E}[e^{\theta X}].$$

We write the expectation as an integral:

$$M_X(\theta) = \int_{-\infty}^{\infty} e^{\theta x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2\theta\sigma^2 x}{2\sigma^2}} dx.$$

By completing the square,

$$x^2 - 2\theta\sigma^2 x = (x - \theta\sigma^2)^2 - \theta^2\sigma^4,$$

we get

$$M_X(\theta) = e^{\frac{\sigma^2\theta^2}{2}} \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{(x-\theta\sigma^2)^2}{2\sigma^2}} dx}_{=1} = e^{\frac{\sigma^2\theta^2}{2}}.$$

Example: Cumulant Generating Function of a Gaussian

Since $K_X(\theta) = \log M_X(\theta)$ and

$$M_X(\theta) = e^{\frac{\sigma^2 \theta^2}{2}},$$

we get

$$K_X(\theta) = \log\left(e^{\frac{\sigma^2 \theta^2}{2}}\right) = \frac{\sigma^2 \theta^2}{2}.$$

Largest Eigenvalue

Proposition 1

Let $\mathbf{X}$ be a random symmetric (Hermitian) matrix. Then

$$\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq \inf_{\theta > 0} \left\{ e^{-\theta t} \cdot \mathbb{E}[\text{tr } e^{\theta \mathbf{X}}] \right\}. \quad (1)$$

Proof. Fix $\theta > 0$. Then

$$\begin{aligned} \Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} &= \Pr\{\lambda_{\max}(\theta \mathbf{X}) \geq \theta t\} \\ &= \Pr\left\{ e^{\lambda_{\max}(\theta \mathbf{X})} \geq e^{\theta t} \right\} \\ &\leq e^{-\theta t} \mathbb{E}\left[e^{\lambda_{\max}(\theta \mathbf{X})} \right], \end{aligned}$$

where the last inequality follows from the **Markov inequality**:

$$\Pr\{Z \geq a\} \leq \frac{\mathbb{E}[Z]}{a}, \quad Z \geq 0, \ a > 0.$$

Largest Eigenvalue

From the previous slide: Given $\theta > 0$, $\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq e^{-\theta t} \mathbb{E} e^{\lambda_{\max}(\theta \mathbf{X})}$

We now use

$$e^{\lambda_{\max}(\theta \mathbf{X})} = \lambda_{\max}(e^{\theta \mathbf{X}}) \leq \text{tr } e^{\theta \mathbf{X}},$$

where the equality is due to the so-called **spectral mapping theorem**¹.

Theorem 2 (spectral mapping theorem)

Let $\mathbf{X}$ be a symmetric square matrix and let $q(x)$ be any polynomial. Then the spectrum of $q(\mathbf{X})$ is the image of the spectrum of $\mathbf{X}$ under q , i.e., $\lambda(q(\mathbf{X})) = q(\lambda(\mathbf{X}))$.

¹To see how to apply such theorem to the exponential matrix, see the following [notes](#)

Largest Eigenvalue

From the previous slide: Given $\theta > 0$, $\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq e^{-\theta t} \mathbb{E} e^{\lambda_{\max}(\theta \mathbf{X})}$

We now use

$$e^{\lambda_{\max}(\theta \mathbf{X})} = \lambda_{\max}(e^{\theta \mathbf{X}}) \leq \operatorname{tr} e^{\theta \mathbf{X}},$$

where the equality is due to the so-called **spectral mapping theorem**¹.

Therefore, we get

$$\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq e^{-\theta t} \mathbb{E} \operatorname{tr} e^{(\theta \mathbf{X})}.$$

Since the left-hand side is **independent** of θ , the infimum over $\theta > 0$ is still an upper bound for it. $\square$

¹To see how to apply such theorem to the exponential matrix, see the following [notes](#)

Lieb's Concavity Theorem: Motivation (1/2)

We just found

$$\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq \inf_{\theta > 0} \{e^{-\theta t} \mathbb{E}[\operatorname{tr} e^{\theta \mathbf{X}}]\}.$$

- For sums

$$\mathbf{X} = \sum_k \mathbf{X}_k,$$

extending scalar concentration bounds to matrices is not trivial:

$$\exp(\mathbf{A} + \mathbf{B}) \neq \exp(\mathbf{A}) \exp(\mathbf{B}) \quad \text{in general.}$$

- This non-commutativity complicates bounding $\mathbb{E}[\operatorname{tr} e^{\theta \mathbf{X}}]$.
- Next we show a simple example illustrating this lack of equality.

Lieb's Concavity Theorem: Motivation (2/2)

Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $A^2 = 0$, so

$$e^A = I + A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad e^B = \text{diag}(e, 1).$$

Meanwhile,

$$e^{A+B} = \begin{pmatrix} e & e-1 \\ 0 & 1 \end{pmatrix}, \quad e^A e^B = \begin{pmatrix} e & 1 \\ 0 & 1 \end{pmatrix}.$$

Clearly $e^{A+B} \neq e^A e^B$.

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Next step

To control $\mathbb{E}[\text{tr} e^{\theta \mathbf{X}}]$, we exploit a result on the **concavity of the trace exponential** – Lieb's theorem.

Lieb's concavity theorem: Statement

Theorem 2 (Concavity property of trace exponential function)

Fix an Hermitian matrix $\mathbf{H}$ with dimension d . The function

$$\mathbf{A} \longmapsto \text{tr}(\exp(\mathbf{H} + \log \mathbf{A})) \text{ is a \textbf{concave map}}$$

on the set of $d \times d$ positive-definite matrices.

Proof omitted, see E.H. Lieb, [Convex trace functions and the Wigner-Yanase-Dyson conjecture](#).

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Meaning

Set

$$f(\mathbf{A}) = \text{tr}(\exp(\mathbf{H} + \log \mathbf{A})).$$

Theorem 2 means: for any two positive-definite matrices $\mathbf{A}, \mathbf{B}$ and any $\tau \in [0, 1]$,

$$f(\tau \mathbf{A} + (1 - \tau) \mathbf{B}) \geq \tau f(\mathbf{A}) + (1 - \tau) f(\mathbf{B}).$$

Corollary of Lieb's Concavity Argument

Corollary 1

Let $\mathbf{H}$ be a fixed Hermitian matrix, and let $\mathbf{Z}$ be a random Hermitian matrix of the same dimension. Then

$$\mathbb{E} \operatorname{tr}(\exp(\mathbf{H} + \mathbf{Z})) \leq \operatorname{tr}(\exp(\mathbf{H} + \log \mathbb{E} e^{\mathbf{Z}})).$$

Proof Corollary

Let $\mathbf{Y} = e^{\mathbf{Z}}$. Then

$$\begin{aligned}\mathbb{E} \operatorname{tr}(\exp(\mathbf{H} + \mathbf{Z})) &= \mathbb{E} \operatorname{tr}(\exp(\mathbf{H} + \log \mathbf{Y})) \\ &\leq \operatorname{tr}(\exp(\mathbf{H} + \log \mathbb{E} \mathbf{Y})) = \operatorname{tr}(\exp(\mathbf{H} + \log \mathbb{E} e^{\mathbf{Z}}))\end{aligned}$$

where

1. The first identity follows from the interpretation of the matrix logarithm as the functional inverse of the matrix exponential ($\log e^{\mathbf{A}} = \mathbf{A}$).
2. Theorem 2 shows that the trace function is concave in $\mathbf{Y}$, so Jensen's inequality² allows us to draw the expectation inside the function.

□

²**Jensen's inequality:** Let $\mathbf{V}$ be a random matrix, and let h be a real-valued function on matrices. Then $\mathbb{E}h(\mathbf{V}) \geq h(\mathbb{E}\mathbf{V})$ when h is convex, and $\mathbb{E}h(\mathbf{V}) \leq h(\mathbb{E}\mathbf{V})$ when h is concave.

Sub-additivity of Matrix Cumulant Generative Functions

Recall: Everything started because we wanted to upper bound the right-hand side of

$$\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq \inf_{\theta > 0} \{e^{-\theta t} \mathbb{E}[\text{tr} e^{\theta \mathbf{X}}]\}$$

when

$$\mathbf{X} = \sum_k \mathbf{X}_k.$$

By applying the concavity property of trace exponential function, we can go further:

Lemma 1 (Subadditivity of Matrix CGFs, in J.A. Tropp, [An Introduction to Matrix Concentration Inequalities](#))

Consider a finite sequence $\{\mathbf{X}_k\}$ of independent, random, Hermitian matrices. Then

$$\mathbb{E} \text{tr} \left(\exp \left(\sum_k \theta \mathbf{X}_k \right) \right) \leq \text{tr} \left(\exp \left(\sum_k \log \mathbb{E} e^{\theta \mathbf{X}_k} \right) \right) \quad \text{for } \theta \in \mathbb{R}.$$

Lemma Proof: $\mathbb{E} \operatorname{tr}(\exp(\sum_k \theta \mathbf{X}_k)) \leq \operatorname{tr}(\exp(\sum_k \log \mathbb{E} e^{\theta \mathbf{X}_k}))$

- W.l.o.g., we assume that $\theta = 1$ by absorbing the parameter into the random matrices.
- Let $\mathbb{E}_k$ be the expectation with respect to $\mathbf{X}_k$, the remaining random matrices held fixed.
- Abbreviate $\Xi_k = \log \mathbb{E}_k e^{\mathbf{X}_k} = \log \mathbb{E} e^{\mathbf{X}_k}$. We may calculate that³

$$\begin{aligned} \mathbb{E} \operatorname{tr} \left(\exp \left(\sum_{k=1}^n \mathbf{X}_k \right) \right) &= \mathbb{E} \mathbb{E}_n \operatorname{tr} \left(\exp \left(\sum_{k=1}^{n-1} \mathbf{X}_k + \mathbf{X}_n \right) \right) \\ &\leq \mathbb{E} \operatorname{tr} \left(\exp \left(\sum_{k=1}^{n-1} \mathbf{X}_k + \log \mathbb{E}_n e^{\mathbf{X}_n} \right) \right) \\ &= \mathbb{E} \mathbb{E}_{n-1} \operatorname{tr} \left(\exp \left(\sum_{k=1}^{n-2} \mathbf{X}_k + \mathbf{X}_{n-1} + \Xi_n \right) \right) \\ &\leq \mathbb{E} \mathbb{E}_{n-2} \operatorname{tr} \left(\exp \left(\sum_{k=1}^{n-2} \mathbf{X}_k + \Xi_{n-1} + \Xi_n \right) \right) \\ &\dots \leq \operatorname{tr} \left(\exp \left(\sum_{k=1}^n \Xi_k \right) \right) \end{aligned}$$

³Iterated expectations are used within the **tower property of conditional expectation**: $\mathbb{E}[\mathbf{X}] = \mathbb{E}_{\mathbf{Y}}[\mathbb{E}_{\mathbf{X}}[\mathbf{X}|\mathbf{Y}]]$

Lemma Proof: $\mathbb{E} \operatorname{tr}(\exp(\sum_k \theta \mathbf{X}_k)) \leq \operatorname{tr}(\exp(\sum_k \log \mathbb{E} e^{\theta \mathbf{X}_k}))$

Note that, in the previous slide, to bound the expectation $\mathbb{E}_m$ for an index $m = 1, 2, 3, \dots, n$, we invoked **Corollary 1**, which stated:

Corollary 1 (see slide 12)

Let $\mathbf{H}$ be a fixed Hermitian matrix, and let $\mathbf{Z}$ be a random Hermitian matrix of the same dimension. Then

$$\mathbb{E} \operatorname{tr}(\exp(\mathbf{H} + \mathbf{Z})) \leq \operatorname{tr}(\exp(\mathbf{H} + \log \mathbb{E} e^{\mathbf{Z}})).$$

For $\mathbf{H}$ equal to

$$\mathbf{H}_m = \sum_{k=1}^{m-1} \mathbf{X}_k + \sum_{k=m+1}^n \mathbf{\Xi}_k,$$

and $\mathbf{Z} = \mathbf{X}_m$, the use of the Corollary is legitimate because $\mathbf{H}_m$ is independent of $\mathbf{X}_m$. □

What's next

Recall: The goal was to upper bound the right-hand side of

$$\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq \inf_{\theta > 0} \{e^{-\theta t} \mathbb{E}[\operatorname{tr} e^{\theta \mathbf{X}}]\}$$

when $\mathbf{X} = \sum_k \mathbf{X}_k$. We proved that

$$\mathbb{E}[\operatorname{tr} e^{\sum_k \theta \mathbf{X}_k}] \leq \operatorname{tr} \left(\exp \left(\sum_k \log \mathbb{E} e^{\theta \mathbf{X}_k} \right) \right).$$

Now it remains to bound $\mathbb{E} e^{\theta \mathbf{X}_k}$.

Matrix Bernstein: Mgf and Cgf Bound

We begin with a lemma to bound the MGF and the CGF of a random matrix $\mathbf{Z}$:

Lemma 2 (Matrix Bernstein: MGF and CGF Bound)

Suppose that $\mathbf{Z}$ is a random Hermitian matrix that satisfies

$$\mathbb{E} \mathbf{Z} = \mathbf{0} \quad \text{and} \quad \lambda_{\max}(\mathbf{Z}) \leq \mathbf{L},$$

then, for $0 < \theta < 3/L$,

$$\mathbb{E} e^{\theta \mathbf{Z}} \preceq \exp\left(\frac{\theta^2/2}{1 - \theta L/3} \cdot \mathbb{E} \mathbf{Z}^2\right) \quad \text{and} \quad \log \mathbb{E} e^{\theta \mathbf{Z}} \preceq \frac{\theta^2/2}{1 - \theta L/3} \cdot \mathbb{E} \mathbf{Z}^2.$$

(the expression $\mathbf{Z} \preceq \mathbf{Y}$ means that $\mathbf{Y} - \mathbf{Z}$ is positive semidefinite, i.e. $\mathbf{x}^\top (\mathbf{Y} - \mathbf{Z}) \mathbf{x} \geq 0 \ \forall \mathbf{x}$)

Lemma Proof (1/2)

$$\begin{aligned} e^{\theta \mathbf{Z}} &= \mathbf{I} + \theta \mathbf{Z} + \sum_{p=2}^{\infty} \frac{\theta^p}{p!} \mathbf{Z}^p \\ &\preccurlyeq \mathbf{I} + \theta \mathbf{Z} + \left(\sum_{p=2}^{\infty} \frac{|\theta|^p}{2 \cdot 3^{p-2}} \|\mathbf{Z}\|^{p-2} \right) \cdot \mathbf{Z}^2 \quad (\text{since } p! \geq 2 \cdot 3^{p-2}) \\ &\preccurlyeq \mathbf{I} + \theta \mathbf{Z} + \frac{\theta^2}{2} \left(\sum_{p=2}^{\infty} \left(\frac{|\theta|}{3} \right)^{p-2} \|\mathbf{Z}\|^{p-2} \right) \cdot \mathbf{Z}^2 \\ &\preccurlyeq \mathbf{I} + \theta \mathbf{Z} + \frac{\theta^2}{2} \left(\sum_{p=2}^{\infty} \left(\frac{|\theta|L}{3} \right)^{p-2} \right) \cdot \mathbf{Z}^2 \\ &= \mathbf{I} + \theta \mathbf{Z} + \frac{\theta^2/2}{1 - |\theta|L/3} \cdot \mathbf{Z}^2, \end{aligned}$$

where we used the formula for geometric series in the last equality (requiring $0 < \theta < 3/L$).

Lemma Proof (2/2)

In the previous slide we found:

$$e^{\theta \mathbf{Z}} \preceq \mathbf{I} + \theta \mathbf{Z} + \frac{\theta^2/2}{1 - |\theta|L/3} \cdot \mathbf{Z}^2.$$

We then take the expectation and use the fact that $\mathbf{Z}$ has zero mean:

$$\begin{aligned} \mathbb{E}e^{\theta \mathbf{Z}} &\preceq \mathbb{E}\left(\mathbf{I} + \theta \mathbf{Z} + \frac{\theta^2/2}{1 - |\theta|L/3} \cdot \mathbf{Z}^2\right) \preceq \mathbb{E}\left(\mathbf{I} + \frac{\theta^2/2}{1 - |\theta|L/3} \cdot \mathbf{Z}^2\right) \\ &\preceq \mathbb{E}\left(\frac{\theta^2/2}{1 - |\theta|L/3} \cdot \mathbf{Z}^2\right) \end{aligned}$$

where we invoked the **transfer rule**⁴ to $1 + a \leq e^a$.

For the other inequality, we use the **logarithm is monotone** w.r.t. $\preceq$, see Prop. 8.4.4 in J.A. Tropp, [An Introduction to Matrix Concentration Inequalities](#). □

⁴**Transfer rule:** $f(a) \leq g(a) \forall a \in I \subset \mathbb{R}$ implies $f(\mathbf{A}) \preceq g(\mathbf{A})$ for any $\mathbf{A}$ whose eigenvalues $\in I$.

Matrix Bernstein

We are now ready to state the **analogue of Bernstein inequality for sum of indep. matrices**.

Theorem 3 (Matrix Bernstein)

Consider a finite sequence $\{\mathbf{X}_k\}$ of independent, random, Hermitian matrices with dimension $d \times d$. Assume that each random matrix satisfies

$$\mathbb{E} \mathbf{X}_k = \mathbf{0} \quad \text{and} \quad \lambda_{\max}(\mathbf{X}_k) \leq \mathbf{L} \quad \text{almost surely}$$

then, for all $t \geq 0$,

$$\underbrace{\Pr\left(\lambda_{\max}\left(\sum_k \mathbf{X}_k\right) \geq t\right)}_{\text{exponentially decaying bounds on the probability that the maximum eigenvalue deviates substantially from zero}} \leq d \cdot \exp\left(\frac{-t^2/2}{\sigma^2 + Lt/3}\right)$$

where $\sigma^2 := \|\sum_k \mathbb{E}(\mathbf{X}_k^2)\|$

Theorem Proof (Sketch)

Consider a finite sequence $\{\mathbf{X}_k\}$ of random Hermitian matrices with dimension $d \times d$. Assume that

$$\mathbb{E} \mathbf{X}_k = \mathbf{0} \quad \text{and} \quad \lambda_{\max}(\mathbf{X}_k) \leq \mathbf{L} \quad \text{for each index } k$$

then the matrix Bernstein cgf bound, Lemma 2, provides that

$$\log \mathbb{E} e^{\theta \mathbf{X}_k} \preceq g(\theta) \cdot \mathbb{E} \mathbf{X}_k^2 \quad \text{where} \quad g(\theta) = \frac{\theta^2/2}{1 - \theta L/3} \quad \text{for } 0 < \theta < 3/L. \quad (2)$$

Introduce the sum $\mathbf{X} = \sum_k \mathbf{X}_k$.

Theorem Proof

Using Eq. (1):

$$\Pr\{\lambda_{\max}(\mathbf{X}) \geq t\} \leq \inf_{\theta > 0} \{e^{-\theta t} \cdot \mathbb{E}[\operatorname{tr} e^{\theta \mathbf{X}}]\}$$

and Lemma 1:

$$\mathbb{E} \operatorname{tr} \left(\exp \left(\sum_k \theta \mathbf{X}_k \right) \right) \leq \operatorname{tr} \left(\exp \left(\sum_k \log \mathbb{E} e^{\theta \mathbf{X}_k} \right) \right)$$

we have ($\mathbf{X} = \sum_k \mathbf{X}_k$)

$$\begin{aligned} \Pr(\lambda_{\max}(\mathbf{X}) \geq t) &\leq \inf_{\theta > 0} e^{-\theta t} \operatorname{tr} \left(\exp \left(\sum_k \log \mathbb{E} e^{\theta \mathbf{X}_k} \right) \right) \\ &\stackrel{(i)}{\leq} \inf_{0 < \theta < 3/L} e^{-\theta t} \operatorname{tr} \left(\exp \left(g(\theta) \sum_k \mathbb{E} \mathbf{X}_k^2 \right) \right) \\ &\stackrel{(ii)}{\leq} \inf_{0 < \theta < 3/L} d e^{-\theta t} e^{g(\theta) \cdot \sigma^2}, \end{aligned}$$

where (i) uses Lemma 2 and (ii) uses $\operatorname{tr}(\mathbf{A}) \leq d \lambda_{\max}(\mathbf{A})$ for a matrix $0 \preceq \mathbf{A}$ with dimension $d \times d$.

To conclude we need to plug in the expression for $g(\theta)$ and compute the infimum. □